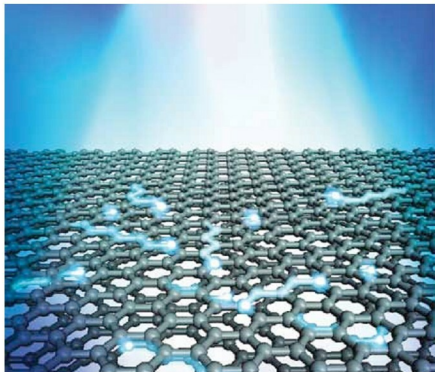




Faster photodetector created by tricking graphene

even absorb very low-energy photons (including terahertz and infrared) which pass through most semiconductors. Interestingly, it has another attractive property as a photon absorber: The electrons which absorb the energy are able to retain it efficiently, rather than losing it to vibrations of the atoms of the material. This property leads to extremely low electrical resistance in graphene.

Researchers exploited these two unique properties of graphene to devise the hot-electron bolometer. The bolometer works by measuring the change in the resistance that results from the heating of the electrons as they absorb light.

Electronic nose to sniff out industrial emissions

Odotech, a spin-out from Polytechnique Montréal, has developed a sensor technology called electronic nose to measure the dispersal of odours from industrial plants into the community. The data collected will enable plant managers to optimise deployment of scrubbing methods and time certain operational processes according to the weather.

"Basically, we have 16 sensors that will react to the same chemical compounds as the human nose does. Based on the reaction of those 16 sensors we generate a fingerprint, so we will need to teach to the electronic nose to be able to recognise different odours," said Thierry Pagé, president at Odotech.

The engineers at Odotech first capture the natural environment near an industrial plant. They also track the odours creating the problems in the labs where they set thresholds. After that, the electronic nose sensor and a micro-weather station are deployed in the immediate vicinity of the plant. The data gathered from these two is processed by Odotech's proprietary software to predict how the odour will spread on an hourly as well as a daily basis. This, in turn, enables plant managers to decide which scrubbers and biofilters are to be used to keep the odour under control.

Invisibility cloak with on-off switch

Physicists from University of St Andrews, Scotland, have created an optical invisibility cloak that you can

turn on and off as you wish. Darran Milne and Natalia Korolkova from the University used the concept of electromagnetically induced transparency in which certain materials become transparent when zapped by light from two carefully tuned lasers.

The cloaking device comprises a transparent mantle that encloses the object to be hidden. The mantle refracts light around its hidden interior such that light exits as if it had traversed empty or uniform space, thus hiding the object and the act of hiding itself.

"This works for materials with atoms that can exist in three different electronic states—say, a, b and, the highest, c. The idea here is that the first laser beam is absorbed by the material because it excites electrons from state 'a' to state 'c'. The second laser is also absorbed because it excites electrons from state 'b' to state 'c'. If the frequencies of the lasers are close together, they can be tuned in a way that makes them interfere destructively. And when this happens, their ability to excite electrons cancels out," Technology Review said in a report.

Open source box to help you create robots from scratch

The Robotic Minion Starter Kit by Dustin and David Andrews on Kickstarter can help you build a robot army if you want a fast start into building robots. For all those interested in robotics, the Robotic Minion Starter Kit is a small computer that includes an LCD, rotary encoder and speaker, all housed in a plastic enclosure.

The easy-to-use Open Source box helps you learn how to control hardware devices such as servo motors, LEDs and relays, which, in turn, can control things like doors, lasers, cameras, music devices and remote control vehicles. The box also performs well for sensor-based projects such as GPS, proximity, temperature, acceleration, air pressure or RFID.

The project is Arduino-compatible, so you can leverage a large base of expertise and code that already exists.

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